Kejun "Albert" Ying, Ph.D.

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Studying aging at the intersection of biology and AI

Education & Experience

Stanford University & University of Washington — Stanford, CA & Seattle, WA
Postdoctoral Researcher — Spring 2025 – Present
- Co-advised by Dr. Tony Wyss-Coray and Dr. David Baker
- NIH/NIA F99/K00 Fellow

Harvard University — Cambridge, MA
Ph.D., Biological Science in Public Health — Fall 2019 – Spring 2025
- Dissertation: "On the Quantification of Aging"
- Advisor: Dr. Vadim Gladyshev, Harvard Medical School, Brigham and Women's Hospital

Harvard University — Cambridge, MA
M.S., Computational Science Engineering — Summer 2023 – Spring 2024
- Secondary field during Ph.D. study

University of California, Berkeley — Berkeley, CA
Visiting Student, Integrative Biology — Summer 2017 – Fall 2017

Sun Yat-Sen University (Yat-Sen Honor School) — Guangzhou, China
B.S., Life Science — Fall 2015 – Summer 2019
- Thesis: Screening for the Interactome of hTERC based on Molecular Fluorescence Complementation System in Living Cells
- Advisor: Dr. Zhou Songyang

Other Professional Experience

Stanford University & University of Washington — Stanford, CA & Seattle, WA
Visiting Scholar, Wyss-Coray's Lab & Baker's Lab — Nov 2024 – May 2025

Harvard Medical School, Boston Children's Hospital — Boston, MA
RNA Modifications *Graduate Researcher (Rotation), Eric Greer's Lab* — Jan 2020 – Mar 2020

Harvard Medical School — Boston, MA
Cell Reprogramming *Graduate Researcher (Rotation), David Sinclair's Lab* — Oct 2019 – Dec 2019

Harvard T. H. Chan School of Public Health — Boston, MA
mTORC1 *Graduate Researcher (Rotation), Brendan Manning's Lab* — July 2019 – Oct 2019

Undergraduate Research — 2015 – 2019

- **Sun Yat-Sen University**, *Telomere & Telomerase* — Zhou Songyang's Lab
- **University of Edinburgh**, *Population genetics* — Xia Shen's Lab
- **University of Washington**, *Acarbose & Raparycin* — Matt Kaeberlein's Lab
- **Buck Institute for Research on Aging**, *Senolytics* — Judith Campisi's Lab
- **University of California, Berkeley**, *SIRT7* — Danica Chen's Lab

- **Sun Yat-Sen University**, *Telomere & DNA Methylation* Yikang Rong's Lab

Grants

THRIVE: Transforming Health: Reclaiming Intrinsic Vitality for Everyone ARPA-H PROSPR

Developing PROSPR Intrinsic Capacity score for predicting 20-year health outcomes, 2026 – 2031

- USD $36 million over 5 years (Key Personnel)
- Multi-institutional collaboration: Stanford University, Harvard University, Buck Institute for Research on Aging
- Integrating multi-omics biomarkers and functional assessments across 19+ longitudinal databases to enable personalized health monitoring and accelerate clinical trials

Aging and Longevity Award ResearchHub Foundation RFP

Funded proposal selected through open peer review, 2026 – 2027

OpenAI Researcher Access Program OpenAI

Research credit grant supporting AI-driven aging biology and protein design, 2026

- USD $20,000 in API credits

Koo Post-doctoral Transition Award NIH/NIA

Using causal aging biomarkers and protein design to develop novel anti-aging interventions, 2025 – 2028

F99 Transition to Aging Research for Predoctoral Students NIH/NIA *Using causal aging biomarkers and protein design to develop novel anti-aging interventions*, 2024 – 2025

- Award Document Number: FAG088431A (PI)
- Received a *perfect* Impact Score of **10**

Publications

† Corresponding author; * Co-first author; + Contributed as consortium author

Selected Publications & Preprints

Wang, Z., Zhang, O., Wang, J., Jiang, Y., Ma, Z. C., ..., Gladyshev, V. N., Wyss-Coray, T., & **Ying, K.**† (2026). A world model simulates the latent dynamics of human health. **medRxiv**. https://doi.org/10.64898/2026.09.19.26363460

Ying, K., Paulson, S., Reinhard, J., ..., Gladyshev, V. N. (2026). An Open Competition for Biomarkers of Aging. **Nature Aging**, *6*, 1193–1195. https://doi.org/10.1038/s43587-026-01139-6

Moqri, M.*, **Ying, K.***, Poganik, J.*, Herzog, C.*, ..., Marioni, R. E., Lasky-Su, J. A., Snyder, M. P., & Gladyshev, V. N. (2026). Integrative epigenetics and transcriptomics identify aging genes in human blood. **Nature Communications**, *16*, 67369. https://doi.org/10.1038/s41467-025-67369-1

Ying, K., Paulson, S., Eames, A., Tyshkovskiy, A., ..., Gladyshev, V. N. (2025). *A Unified Framework for Systematic Curation and Evaluation of Aging Biomarkers*. **Nature Aging**. https://www.nature.com/articles/s43587-025-00987-y

Wu, X.*, Liu, H.*, **Ying, K.***† (2025). Biological Age, Aging Clocks, and the Interplay with Lymphoid Neoplasms: Mechanisms and Clinical Frontiers. **Lymphatics**, *3*(3), 19. https://doi.org/10.3390/lymphatics3030019

Ying, K.† (2024). Causal inference for epigenetic ageing. **Nature Reviews Genetics**, 1–1. https://doi.org/10.1038/s41576-024-00799-7

Ying, K., Castro, J. P., Shindyapina, A. V., ..., Gladyshev, V. N. (2024). Depletion of loss-of-function germline mutations in centenarians reveals longevity genes. **Nature Communications**, *15*(1), 5956. https://doi.org/10.1038/s41467-024-52967-2

Ying, K., Liu, H., Tarkhov, A. E., ..., Gladyshev, V. N. (2024). Causality-enriched epigenetic age uncouples damage and adaptation. **Nature Aging (February Cover)**, 1–16. https://doi.org/10.1038/s43587-023-00557-0

Ying, K., Zhai, R., Pyrkov, T. V., ..., Gladyshev, V. N. (2021). Genetic and phenotypic analysis of the causal relationship between aging and COVID-19. **Communications Medicine**, *1*(1), 35. https://doi.org/10.1038/s43856-021-00033-z

Ying, K.,*† Tyshkovskiy, A., Moldakozhayev, A., ..., Gladyshev, V. N. (2025). Autonomous AI Agents Discover Aging Interventions from Millions of Molecular Profiles. **bioRxiv (Nature under review)**. https://doi.org/10.1101/2023.02.28.530532

Ying, K.†**,** Song, J., Cui, H., ..., Gladyshev, V. N†. (2024). MethylGPT: a foundation model for the DNA methylome. **bioRxiv (Nature Aging 1st Revision)**. https://doi.org/10.1101/2024.10.30.621013

Ying, K., Tyshkovskiy, A., Chen, Q., ..., Gladyshev, V. N. (2024). High-dimensional Ageome Representations of Biological Aging across Functional Modules. **bioRxiv (Nature Aging 2nd Revision)**. https://doi.org/10.1101/2024.09.21.570935

Other Publications

Ying, K.†, Tyshkovskiy, A., Moqri, M., Poganik, J. R., Horie, S., Han, J.-D. J., Ng, C., Gladyshev, V. N., Wyss-Coray, T., & Huang, Y. (2026). Unified frameworks for aging biomarkers: bridging discovery science and clinical translation. In *Toward healthy longevity: Integrating biology, AI, and clinical innovation* (pp. 26–28). **Science/AAAS Custom Publishing Office**. https://www.science.org/content/resource/toward-healthy-longevity-integrating-biology-ai-and-clinical-innovation

Zhavoronkov, A., Galkin, F., Chen, S., ..., **Ying, K.**, ..., Gladyshev, V. N. (2026). Integration of proteomic aging clocks in a phase 2a clinical trial supports simultaneous geroprotective assessment. **Nature Biotechnology**. https://doi.org/10.1038/s41587-026-03286-y

Kordowitzki, P., & **Ying, K.** (2026). Novel Exploratory Transcriptomic Candidates as Biomarkers and Cancer Hallmark Fingerprints for Ovarian Endometroid and Clear Cell Carcinomas in Women. **Antioxidants**, *15*(8), 979. https://doi.org/10.3390/antiox15080979

Wu, F., Xuan, W., Qi, H., ..., **Ying, K.,** ..., Choi, Y. (2026). Proteo-R1: Reasoning Foundation Models for De Novo Protein Design. **ICML 2026**. https://arxiv.org/abs/2605.02937

Zhavoronkov, A., **Ying, K.,** & Wilczok, D. (2026). Peakspan: Defining, Quantifying and Extending the Boundaries of Peak Productive Lifespan. **Aging and Disease**. https://doi.org/10.14336/AD.2026.0080

Kordowitzki, P., & **Ying, K.** (2026). The pursuit of understanding human longevity. **npj Aging**, *12*(1), 25. https://doi.org/10.1038/s41514-026-00339-z

Tyshkovskiy, A., Kholdina, D., **Ying, K.,** Davitadze, M., ..., Gladyshev, V. N. (2026). Universal transcriptomic hallmarks of mammalian ageing and mortality. **Nature**, *654*, 173–188. https://doi.org/10.1038/s41586-026-10542-3

Mavrommatis, C., Belsky, D. W., **Ying, K.,** Moqri, M., Campbell, A., Richmond, A., Gladyshev, V. N., Chandra, T., McCartney, D. L., & Marioni, R. E. (2025). An unbiased comparison of 14 epigenetic clocks in relation to 174

incident disease outcomes. **Nature Communications**, *16*, 11164. https://doi.org/10.1038/s41467-025-66106-y

Zhang, O., Lin, H., Zhang, X., Wang, X., Wu, Z., Ye, Q., Zhao, W., Wang, J., **Ying, K.,**, Kang, Y., Hsieh, C.-Y., Hou, T. (2025). Graph neural networks in modern AI-aided drug discovery. **Chemical Reviews**, 125, 10001–10103. https://doi.org/10.1021/acs.chemrev.5b00254

Zhang, O., ..., **Ying, K.,** Huang, Y., Zhao, H., Kang, Y., Pan, P., Wang, J., Guo, D., Zheng, S., Hsieh, C.-Y., & Hou, T. (2025). ECloudGen: leveraging electron clouds as a latent variable to scale up structure-based molecular design. **Nature Computational Science**. https://doi.org/10.1038/s43588-025-00886-7

Farinas, A., Rutledge, J., Bot, V. A., Western, D., **Ying, K.,** Lawrence, K. A., Oh, H. S. H., ..., Wyss-Coray, T. (2025). Disruption of the cerebrospinal fluid–plasma protein balance in cognitive impairment and aging. **Nature Medicine**, 1–12. https://doi.org/10.1038/s41591-025-03831-3

Rothi, M.H., Sarkar, G.C., Haddad, J.A., Mitchell, W., **Ying, K.,** et al. (2025). The 18S rRNA methyltransferase DIMT-1 regulates lifespan in the germline later in life. **Nature Communications**, 16, 6944. https://doi.org/10.1038/s41467-025-62323-7

Grzeczka, A., Iqbal, S., **Ying, K.,** Kordowitzki, P. (2025). Circular RNAs as regulators and biomarkers of mammalian ovarian ageing. **GeroScience**, 1–19. https://doi.org/10.1007/s11357-025-01798-0

Jacques, E., Herzog, C., **Ying, K.,** ... Gladyshev, V. N. (2025). Invigorating discovery and clinical translation of aging biomarkers. **Nature Aging**, 1–5. https://doi.org/10.1038/s43587-025-00838-w

Goeminne, L. J. E., Vladimirova, A., Eames, A., Tyshkovskiy, A., Argentieri, M. A., **Ying, K.,** Moqri, M., & Gladyshev, V. N. (2025). Plasma protein-based organ-specific aging and mortality models unveil diseases as accelerated aging of organismal systems. **Cell Metabolism**, https://doi.org/10.1016/j.cmet.2024.10.005

Gladyshev, V. N., Anderson, B., Barlit, H., ..., **Ying, K.,** Yunes, J., Zhang, B., & Zhavoronkov, A. (2024). Disagreement on foundational principles of biological aging. **PNAS Nexus**, *3*(12), pgae499. https://doi.org/10.1093/pnasnexus/pgae499

Lyu, YX.[*], Fu, Q.[*], Wilczok, D.[*], **Ying, K.**[*], King, A., ..., Bakula, D. (2024). Longevity biotechnology: Bridging AI, biomarkers, geroscience and clinical applications for healthy longevity. **Aging**, *16*(1), 1–25. https://doi.org/10.18632/aging.206135

Biomarkers of Aging Consortium[+], Herzog, C. M. S., Goeminne, L. J. E., Poganik, J. R., ..., Gladyshev, V. N. (2024). Challenges and recommendations for the translation of biomarkers of aging. **Nature Aging**, 1–12. https://doi.org/10.1038/s43587-024-00683-3

Castro, J. P., Shindyapina, A. V., Barbieri, A., **Ying, K.,** ..., Gladyshev, V. N. (2024). Age-associated clonal B cells drive B cell lymphoma in mice. **Nature Aging**, *4*(8), 1–15. https://doi.org/10.1038/s43587-024-00671-7

Moqri, M., ..., de Sena Brandine, G., **Ying, K.,** Tarkhov, A., ..., Sebastiano, V. (2024). PRC2-AgeIndex as a universal biomarker of aging and rejuvenation. **Nature Communications**, *15*(1), 5956. https://doi.org/10.1038/s41467-024-50098-2

Tarkhov, A. E., Lindstrom-Vautrin, T., Zhang, S., **Ying, K.,** Moqri, M., ..., Gladyshev, V. N. (2024). Nature of epigenetic aging from a single-cell perspective. **Nature Aging**, 1–17. https://doi.org/10.1038/s43587-023-00555-2

Moqri, M., Herzog, C., Poganik, J. R., **Ying, K.,**, ... Ferrucci, L. (2024). Validation of biomarkers of aging. **Nature Medicine**, 1–13. https://doi.org/10.1038/s41591-023-02784-9

Griffin, P. T., ..., Kerepesi, C., **Ying, K.,** ..., Sinclair, D. A. (2024). TIME-seq reduces time and cost of DNA methylation measurement for epigenetic clock construction. **Nature Aging**, 1–14. https://doi.org/10.1038/

s43587-023-00555-2

Moqri, M., Herzog, C., Poganik, J. R., **Biomarkers of Aging Consortium**$^{+}$, ... Gladyshev, V. N. (2023). Biomarkers of aging for the identification and evaluation of longevity interventions. **Cell**, *186*(18), 3758–3775. https://doi.org/10.1016/j.cell.2023.08.003

Liberman, N., Rothi, M. H., Gerashchenko, M. V., Zorbas, C., Boulias, K., MacWhinnie, F. G., **Ying, A. K.,** Flood Taylor, A., ..., Greer, E. L. (2023). 18S rRNA methyltransferases DIMT1 and BUD23 drive intergenerational hormesis. **Molecular Cell**, *83*(18), 3268–3282.e7. https://doi.org/10.1016/j.molcel.2023.08.014

Bitto, A., Grillo, A. S., Ito, T. K., Stanaway, I. B., Nguyen, B. M. G., **Ying, K.,** ... Kaeberlein, M. (2023). Acarbose suppresses symptoms of mitochondrial disease in a mouse model of Leigh syndrome. **Nature Metabolism**, *5*(6), 955–967. https://doi.org/10.1038/s42255-023-00815-w

Emmrich, S., Trapp, A., Tolibzoda Zakusilo, F., Straight, M. E., **Ying, A. K.,** Tyshkovskiy, A., ..., Gorbunova, V. (2022). Characterization of naked mole-rat hematopoiesis reveals unique stem and progenitor cell patterns and neotenic traits. **The EMBO Journal**, *41*(15), e109694. https://doi.org/10.15252/embj.2021109694

Yang, Z., ..., Guo, H., **Ying, K.,** Gustafsson, S., ..., Shen, X. (2022). Genetic Landscape of the ACE2 Coronavirus Receptor. **Circulation**, *145*(18), 1398–1411. https://doi.org/10.1161/CIRCULATIONAHA.121.057888

Li, T., Ning, Z., Yang, Z., Zhai, R., Zheng, C., Xu, W., Wang, Y., **Ying, K.,** Chen, Y., & Shen, X. (2021). Total genetic contribution assessment across the human genome. **Nature Communications**, *12*(1), 2845. https://doi.org/10.1038/s41467-021-23124-w

Zhu, J., Xu, M., Liu, Y., Zhuang, L., **Ying, K.,** Liu, F., ..., Songyang, Z. (2019). Phosphorylation of PLIN3 by AMPK promotes dispersion of lipid droplets during starvation. **Protein & Cell**, *10*(5), 382–387. https://doi.org/10.1007/s13238-018-0593-9

Other Preprints

Cui, T., Wang, X., Li, Z., Ding, J., ..., **Ying, K.,** ..., Wu, Y. (2026). An immune world model for multiscale forecasting and therapeutic hypothesis generation. **arXiv**. arXiv:2609.14709

Zhou, Y., Wang, Y., Wang, J., Zhu, H., ..., **Ying, K.,** ..., Zhang, O. (2026). ADMET-EvO: a self-evolving scientific agent for sustained research across heterogeneous tasks. **arXiv**. arXiv:2609.10121

Jiang, S., Fang, Z., ..., **Ying, K.,** ..., Ding, J. (2026). The Landscape of Single-Cell Foundation Models: Design Principles, Applications, and Open Challenges. **Preprints.org**. doi:10.20944/preprints202608.1166.v1

Xi, Z., Zhou, E., Fang, X., ..., **Ying, K.,** ..., Huang, X. (2026). AI for Productivity in the Age of Agentic AI. **Preprints.org**. doi:10.20944/preprints202608.1248.v1

Zhang, S., Tyshkovskiy, A., **Ying, K.,** Wang, S., & Gladyshev, V. N. (2026). Mammalian aging involves genome-wide splicing degeneration leading to functional decline. **bioRxiv**. https://doi.org/10.64898/2026.06.26.734787

Eames, A., Glubokov, D., Moldakozhayev, A., Yücel, A. D., Tyshkovskiy, A., **Ying, K.,** Goeminne, L. J. E., de Magalhães, C. G., & Gladyshev, V. N. (2026). Anatomy of aging through organ-resolved multi-modal imaging and deep learning. **medRxiv**. https://doi.org/10.64898/2026.03.14.26348392

Roberts, K. F., Abrams, Z. B., Cappelletti, L., Moqri, M., ..., **Ying, K.,** ..., Reese, J. T. (2026). OpenScientist: evaluating an open agentic AI co-scientist to accelerate biomedical discovery. **medRxiv**. https://doi.org/10.64898/2026.03.15.26348338

Wu, Y. C., Yin, M., Shi, B., Zhang, Z., ..., **Ying, K. A.,** ..., Wang, M., & Cong, L. (2026). MedOS: AI-XR-Cobot World Model for Clinical Perception and Action. **medRxiv (withdrawn)**. https://doi.org/10.64898/2026.02.18.26345936

Moldakozhayev, A., Tyshkovskiy, A., Nigro, P., de Magalhães, C. G., **Ying, K.,** ..., Gladyshev, V. N. (2025). Rejuvenation of white adipose tissue in a longitudinal heterochronic transplantation model. **bioRxiv**. doi:10.64898/2025.12.28.696721

Zhang, O., Zhang, X., Lin, H., Tan, C., Wang, Q., Mo, Y., ..., **Ying, K.,** Li, J., Zeng, Y., Lang, L., Pan, P., Cao, H., Song, Z., Qiang, B., Wang, J., Ji, P., Bai, L., Zhang, J., Hsieh, C.-Y., Heng, P. A., Sun, S., Hou, T., & Zheng, S. (2025). ODesign: A World Model for Biomolecular Interaction Design. **arXiv**. https://odesign1.github.io/

Mavrommatis, C., Belsky, D., **Ying, K.,** Moqri, M., Campbell, A., Richmond, A., ..., Gladyshev, V. N. (2025). *An unbiased comparison of 14 epigenetic clocks in relation to 10-year onset of 174 disease outcomes in 18,859 individuals*. **medRxiv**. https://doi.org/10.1101/2025.07.14.25331494

Galkin, F., ..., Tyshkovskiy, A., **Ying, K.,** Gladyshev, V. N., & Zhavoronkov, A. (2024). Precious3GPT: Multimodal Multi-Species Multi-Omics Multi-Tissue Transformer for Aging Research and Drug Discovery. **bioRxiv**. https://doi.org/10.1101/2024.07.25.605062

Rothi, M. H., Sarkar, G. C., Al Haddad, J., Mitchell, W., **Ying, K.,** Pohl, N., ..., Greer, E. L. (2024). The 18S rRNA Methyltransferase DIMT-1 Regulates Lifespan in the Germline Later in Life. **bioRxiv**. https://doi.org/10.1101/2024.05.15.570935

Zhang, B., Tarkhov, A. E., Ratzan, W., **Ying, K.,** Moqri, M., ..., Gladyshev, V. N. (2022). *Epigenetic profiling and incidence of disrupted development point to gastrulation as aging ground zero in Xenopus laevis*. **bioRxiv**. https://doi.org/10.1101/2022.08.02.502559

Patents

V. N. Gladyshev, **K. Ying**, "High-dimensional measurement of biologicar age" (2024). *Provisional Patent Application*

V. N. Gladyshev, **K. Ying**, "Mapping CpG sites to quantify aging traits" (2024). *WO2024039905A2*

Software and Database

Autonomous Lab (2026) https://github.com/albert-ying/autonomous-lab
MCP server for multi-agent autonomous workflows; AI agents execute senior-junior loops while humans steer decisions.

ClockBase Agent (2025) https://www.clockbase.org/
AI agent system that autonomously discovers aging-modifying interventions from ~2M human and mouse molecular profiles.

MethylGPT (2024) https://github.com/albert-ying/MethylGPT
Foundation model for the DNA methylome, pretrained on 226K+ methylation profiles from 5,281 datasets.

Biolearn (2024) https://bio-learn.github.io/
Open-source Python library for biomarkers of aging analysis with reference implementations of common aging clocks.

ClockBase (2023) http://gladyshevlab.org:3838/ClockBase/
Comprehensive platform for biological age profiling across multiple clocks in human and mouse (~300K samples).

Presentations

Oral Presentations

Biomarkers of Aging Symposium 2025 — Boston, MA
Massive AI agent mining of aging-modifying interventions from millions of molecular profile — 2025

ASHG 2025 Annual Meeting — Boston, MA
Decoding the Aging Methylome: From Causal Inference to Foundation Models — 2025

6th TimePie Longevity Forum — Shanghai, China
Massive AI agent mining of aging-modifying interventions from millions of molecular profile — 2025

CSH-Asia Conference: Stem cell, Aging and Rejuvenation — Suzhou, China
Massive AI agent mining of aging-modifying interventions from millions of molecular profile — 2025

Keystone Symposia: Aging: New Frontiers in Rejuvenation and Gerotherapeutics — Breckenridge, CO
MethylGPT: A Foundation Model for the DNA Methylome — 2025

Biomarkers of Aging Symposium — Boston, MA
Standardization of aging biomarkers and BoA challenge — 2024

Harvard GRIP Presentations — Boston, MA
Causal Aging Biomarker empowers Unbiased Anti-Aging Therapy Screening — 2024

4th TimePie Longevity Forum — Shanghai, China
Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening — 2023

Global Congress on Aesthetic and Anti-Aging (GCAA2023) — Singapore
Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening — 2023

10th Aging Research and Drug Discovery conference (ARDD2023) — Copenhagen, Denmark
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

AGE 2023 51st Annual Meeting — Oklahoma City, OK
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Broad Institute MPG Retreat — Cambridge, MA
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Harvard GRIP Presentations — Boston, MA
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

Targeting Metabesity 2022, 'Honorable Mention' — Virtual Conference
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

GSA 2021 Annual Scientific Meeting — Virtual Conference
Genetic and phenotypic evidence for causal relationships between aging and COVID-19 — 2021

Invited Talks

St. Jude Children's Research Hospital, hosted by Dr. Zhaoming Wang — Memphis, TN
MethylGPT and Causality-enriched Epigenetic Clock — 2025

The Alliance for Longevity Initiatives Scientist Spotlight, — Online Podcast
Episode 14: Albert Ying — 2024

BioAge Seminar, hosted by Dr. Robert Hughes & Dr. Paul Timmers — Boston, MA
Ageome: Biological age with higher-dimensionality — 2024

MRC Integrative Epidemiology Unit Seminar — Bristol, UK
Epigenetic Clocks and Mendelian Randomization — 2024

NIA EL Projects Joint Meeting, National Institute on Aging — Online Webinar
Aging Clocks — 2024

Biomarkers of Aging Challenge, Foresight Institute — Online Webinar
Update Webinar with Foresight — 2024

Everything Epigenetics, podcast hosted by Hannah Went — Online Podcast
Causal Epigenetic Age Uncouples Damage and Adaptation — 2024

Chinese University of Hong Kong, hosted by Dr. Xin Wang — Hong Kong, China
Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening — 2024

Everything Epigenetics, podcast hosted by Hannah Went — Online Podcast
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Chinese University of Hong Kong, hosted by Dr. Xin Wang — Hong Kong, China
Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening — 2023

Peking University, hosted by Dr. Jingdong Han — Beijing, China
Causal Aging Biomarker and ClockBase — 2023

Chinese Academy of Sciences, hosted by Dr. Xuming Zhou — Beijing, China
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

Foresight Institute, hosted by Allison Duettmann — Online Seminar
Genetic Variation, Aging & Relationship to COVID-19 | Joris Deelen, Albert Ying — 2020

Honors

WAIC Yunfan Award, "Tomorrow Star", World Artificial Intelligence Conference — 2026

Butler-Williams Scholar (Selected but unable to attend), National Institute on Aging (NIH) — 2026

Semifinalist, Harvard President's Innovation Challenge, Health Care and Life Sciences Track — 2025

Best Poster Award, Inaugural Biomarker of Aging Symposium — 2023

Best Poster Award, Gordon Research Conference, Systems Aging — 2022

Hackathon Winner, Longevity Hackathon, VitaDAO — 2021

Yat-Sen Honor School Program, Sun Yat-Sen University — 2016 – 2019

Yat-Sen Scholarship, Sun Yat-Sen University — 2016 – 2019

Service & Leadership

Board Member, Lifeboat Foundation — 2025 – Present

Member, Norn Longevity Nexus — 2025 – Present

Jury and Mentor, Agentic AI Against Aging Hackathon 2025

Core Member, Biomarkers of Aging Consortium 2024 – Present

Organizer, Biomarker of Aging Challenge 2024 – Present

President, Harvard Interdisciplinary Discussion on Disease and Health 2024 – 2025

Agenda Contributor, World Economic Forum 2024

Organizing Committee Member, Biomarker of Aging Symposium 2024 2024

Organizing Committee Member, Biomarker of Aging Symposium 2023 2023

Teaching & Mentoring

Mentor, Yuanpei Young Scholars Program 2023 – 2024

Instructor, Harvard Public Health Symposium For Young Generation 2023

Students Supervised

Predoctoral Students: Ali Doga Yucel, Siyuan Li, Hanna Liu, Donghyun Lee, Yikun Zhang

Journals Reviewed

Nature, *Nature Aging*, *Nature Communications*, *Communications Medicine*, *Genome Medicine*, *BMC Nephrology*, *Scientific Reports*, ...

Selected Press & Media

Science — Eric Topol, "Learning the language of life with AI" Jan 2025
Featuring MethylGPT.

Nature Aging — "Epigenetic clock work ticks forward" Feb 2024
Expert commentary featuring causal epigenetic clocks.

Harvard Gazette — "Looking to rewind the aging clock" Feb 2024
Featuring causal epigenetic clocks.

World Economic Forum — "Not just a number: Science is redefining how we understand ageing" Nov 2024
Featuring aging clocks for global policy audience.

Nature Biotechnology — "Scientists hone tools to measure aging and rejuvenation interventions" Aug 2023
Featuring ClockBase platform for aging intervention screening.

Lifespan.io — "An AI-Based System Has Found a Potential Longevity Drug" Dec 2025
Featuring ClockBase Agent and ouabain discovery.

MIT Technology Review — "A 1990s-Born Chinese Scientist Sparks Anti-Aging Breakthrough" Dec 2025
Featuring AI agent-driven aging intervention discovery.

PÚBLICO (Portugal's leading newspaper) — "O que dizem os genes de quem tem 100 anos" Dec 2024 Featuring centenarian longevity genes.

Diário de Notícias (Portugal) — "Que segredos escondem os genes de quem vive até aos 100 anos?" Dec 2024

ScienceDaily — "New epigenetic clocks reinvent how we measure age" Feb 2024

References

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